$$\begin{aligned}\mathcal{C}_{\rm osc}(\tau) &= \int \mathrm{d}\omega\, J_{\rm osc}(\omega) \left\{ \coth\left(\frac{\omega}{2k_{\rm B}T}\right) \cos(\omega\tau) - \mathrm{i}\sin(\omega\tau) \right\} \\ &\equiv \nu(\tau) - \mathrm{i}\eta(\tau), \end{aligned} \tag{5.187}$$